



CRITIC-TOPSIS Method Based on Pairwise Preference Graph Neural Similarity and Its Application in Vehicle Selection

Jianping Fan¹ · Yanlong Han¹ · Meiqin Wu¹

Received: 23 August 2024 / Revised: 7 October 2024 / Accepted: 12 December 2024 / Published online: 18 February 2025
© The Author(s) under exclusive licence to Taiwan Fuzzy Systems Association 2025

Abstract In recent years, the rise and rapid development of new energy vehicles (NEVs) has driven the global vehicle industry to transition from traditional fuel-powered models to electric vehicles. The selection of electric vehicles involves uncertainty and complexity and can be viewed as a complex decision problem. Therefore, based on the pairwise preference graph neural similarity (PPGNS) calculation method, this paper proposes a decision framework for addressing vehicle selection problems. First, we represent each expert's pairwise preferences as graphs and use a graph similarity neural network to predict the similarity between these graphs, assigning expert weights based on the similarity levels. Next, we introduce the Pythagorean fuzzy weighted arithmetic mean (PFWAM) aggregation operator to combine expert weights and individual evaluation information to obtain the group decision matrix. Then, we determine attribute importance using the criteria importance through inter-criteria correlation (CRITIC) method in a Pythagorean fuzzy environment and rank the alternatives using the extended TOPSIS. Finally, sensitivity and comparative analyses confirm the effectiveness and practicality of the proposed method in vehicle selection problems.

Keywords MAGDM · Pythagorean fuzzy sets · Graph neural networks · CRITIC · TOPSIS

1 Introduction

With the rapid development of the global automotive industry, cars have become a part of people's daily lives, and the number of cars is growing exponentially. The environmental pollution and energy crisis caused by traditional fuel vehicles are becoming increasingly prominent [1]. The emissions of carbon dioxide and other gases from conventional fuel vehicles will not only worsen the atmospheric environment, but also affect people's health. Therefore, the promotion of NEVs has become an important measure to deal with environmental and ecological problems in the world. In particular, the use of NEVs in the taxi fleet instead of conventional gasoline vehicles can improve air quality as taxis often travel in densely populated areas. At the same time, electric vehicles have attracted many taxi drivers because electricity is more affordable and maintenance costs are lower [2]. Because of these aims, the development of the NEVs industry has become an important part of the global strategy to drive innovation in the fight against climate change [3]. Using China as an example, numerous enterprises dedicated to enhance the performance of NEVs have emerged, such as BYD, NIO, Xpeng Motors, developing a variety of models with different specifications and capabilities of NEVs. Although all companies have made great contributions to the innovation of NEVs, it has also made choosing vehicles difficult. In recent years, many scholars have studied various models and methods for selecting NEVs [4–7], as choosing the right vehicle is crucial for drivers. Profit-oriented drivers prefer NEVs that are cheap and perform better. Since different drivers have different preferences for cars, including appearance, performance, and other parameters, these preferences are vague and uncertain. Therefore, developing a Multi-attribute group decision-making (MAGDM) method based on preferences in

✉ Meiqin Wu
wmq80@sxu.edu.cn

Jianping Fan
fjp@sxu.edu.cn

Yanlong Han
202321609009@email.sxu.edu.cn

¹ School of Economics and Management, Shanxi University, Taiyuan 030006, Shanxi, China

a complex and uncertain environment is crucial for solving the problem of vehicle selection.

MAGDM usually involves multiple decision-makers (DMs) ranking limited options based on several criteria to select the best alternative. It is widely used in photovoltaic power station site selection [8], evaluate the healthcare supply chain [9], health management center selection [10], medical health resources allocation evaluation [11], financial logistics enterprise evaluation [12], and other scenarios. In the evaluation process, experts from different fields are usually gathered, and their assessment information is used for decision-making. However, due to the fuzziness and uncertainty of the evaluators' information, making accurate decisions in real situations becomes very difficult. To better describe fuzzy information, Zadeh [13] proposed the concept of fuzzy sets (FS) theory, DMs can use membership grades (MG) in this theory to express their degree of preference. Subsequently, Atanassov [14] extended FS by adding non-membership grades (NMG) and proposed the concept of intuitionistic fuzzy sets (IFS). IFS states that the sum of MG and NMG must be less than or equal to 1. However, not all situations meet this condition. Therefore, Yager [15] proposed the concept of Pythagorean fuzzy sets (PyFS), which also have MG and NMG. The difference is that their squared sums are less than or equal to 1. Therefore, PyFS has become a more effective tool for handling uncertain information.

At the beginning of the MAGDM, due to the differences in DMs' educational backgrounds, knowledge levels, and social relationships, weights should be flexibly assigned to DMs when aggregating their evaluation information. In existing studies, Lin et al. [16] proposed a multi-criteria decision-making model based on the TOPSIS method to evaluate the security of mining systems. In the model, the weights of experts are determined by an expert coefficient. This coefficient considers the expert's judgment ability, subjective reliability, and risk preference. Krishankumar et al. [17] determined the weight of experts through a regret factor. This method considers the hesitation of experts in the preference expression process and provides a reasonable evaluation of expert weights. Wang et al. [18] constructed an optimal model based on similarity to calculate each expert's weight by comparing the deviation between the expert evaluation value and the average value. However, in calculating expert weights, different similarity distance measures, such as the traditional euclidean and cosine distances, are often involved. However, graph similarity measures are rarely used. Consequently, we developed a new method using graph neural networks to predict similarity based on pairwise user preferences for calculating expert weights.

A graph is a mathematical model that represents relationships using nodes and edges. It is widely used in various fields, including social network analysis, recommendation systems, link prediction, and community detection.

In the field of decision science, Yu and Xu [19] first introduced graph structures into Multi-attribute decision-making (MCDM), represented the relationships between criteria with graphs, and constructed a graph-based multi-agent decision-making model. Akram et al. [20] explored the advantages of Pythagorean fuzzy graphs and used them to address fuzziness and uncertainty in network problems. Naz and Akram [21] proposed a graph-based decision-making method and extended it to handle interval-valued hesitant fuzzy and hesitant triangular fuzzy information. Later, Luqman et al. [22] analyzed risks in failure mode and effects analysis using directed graphs and matrix techniques in a Pythagorean fuzzy environment. Meng et al. [23] introduced graph neural networks (GNNs) into MAGDM and developed a Pythagorean fuzzy MAGDM method based on GNNs.

Next, the task is to determine the weights of the criteria and rank the alternatives. For MAGDM problems, attribute weights often have a significant impact on the results [24]. In recent years, various scholars have proposed many methods for calculating attribute weights. These methods are generally divided into two categories: objective methods and subjective methods [25]. Diakoulaki et al. [26] proposed the CRITIC method to calculate attribute weights. Due to its superior performance in objective calculations, it has been well-received and widely used by scholars. Haktanır and Kahraman [27] applied the CRITIC method in Picture fuzzy sets (PFSs) and proposed an integrated CRITIC & REGIME methodology. Negara et al. [28] designed an MCDM-based caching placement strategy that uses the CoCoSo and CRITIC weighting methods to enhance content distribution. Farid et al. [29] designed the CRITIC-EDAS method based on q-rung orthopair fuzzy (q-ROF) for intelligent waste management. In terms of ranking alternatives, the TOPSIS method [30–34] has received considerable attention from researchers for finding compromise solutions in MAGDM problems and has been successfully applied various fields. Therefore, inspired by GNNs and the CRITIC-TOPSIS method, we designed a CRITIC-TOPSIS framework based on pairwise preference graph neural networks in a Pythagorean fuzzy environment to address MAGDM problems.

The research motivations of this paper mainly include: (1) The environmental pollution and energy crisis caused by traditional fuel vehicles are becoming increasingly severe, making the promotion of NEVs a key measure in addressing environmental issues [35]. (2) The existing methods for calculating expert weights largely rely on traditional similarity measures, such as cosine distance [23] and euclidean distance [36], which are typically based on input vectors. However, relatively few studies focus on graph-based structural similarity. (3) Considering that choosing the right electric vehicle involves uncertainty and complexity, it is necessary to develop a MAGDM method to address this issue.

The main contributions of our research are as follows:

- We proposed a rapid graph similarity neural network method based on pairwise preferences to obtain expert weights, converting expert preferences into graphical format to evaluate the similarity between each pair of experts. Prior to this, we introduced Top-k similarity to select the Top k most similar experts as training labels for the model.
- Based on the rapid graph similarity neural network, we developed the PyFS-PPGNS-CRITIC-TOPSIS integrated decision framework to effectively address MAGDM problems. SimGNN was used to deeply mine information from expert preference graphs and predict the similarity matrix between experts, thereby effectively assigning weights to experts. Finally, the PyFS-CRITIC-TOPSIS method was employed to calculate attribute weights and rank the alternatives.
- We introduced the PFWAM operator to generate the decision matrix, effectively overcoming the issue of being unable to distinguish the ranking order of alternatives in certain situations.
- The method was applied to the vehicle selection problem and subjected to sensitivity and comparative analyses, validating its effectiveness and practicality.

The remainder of the paper is arranged as follows: In Sect. 2, we review the necessary definitions, including PyFS and the calculation of graph similarity. In Sect. 3, we develop the PyFS-PPGNS-CRITIC-TOPSIS method to address the vehicle selection problem. Section 4 provides numerical examples applied to the vehicle selection problem, demonstrating the practicality of the method. In Sect. 5, we conduct sensitivity and comparative analyses of the proposed framework to validate its effectiveness. Finally, Sect. 6 summarizes the work and discusses future research directions.

2 Preliminaries

In this section, we briefly introduce some basic definitions of PyFS and some basic concepts of similarity graph neural network (SimGNN).

2.1 Pythagorean Fuzzy Sets

Definition 1 [15] Let X be a finite universe of discourse, than a PyFS P in X is given by

$$A = \{\langle X, \mu(x), \eta(x) \rangle | x \in X\} \quad (1)$$

where $\mu_A : X \rightarrow [0, 1]$, $\eta_A : X \rightarrow [0, 1]$ denote the degree of membership and non-membership, respectively, and their

membership functions satisfy $0 \leq \mu_A^2(X) + \eta_A^2(X) \leq 1$.

The hesitation is $\nu_A(x) = \sqrt{1 - \mu_A^2(X) - \eta_A^2(X)}$. For convenience, the $\kappa = (\mu_\kappa, \eta_\kappa)$ referred to as Pythagorean fuzzy numbers (PFN), where $\mu_\kappa, \eta_\kappa \in [0, 1]$, and $0 \leq \mu_\kappa^2 + \eta_\kappa^2 \leq 1$.

Definition 2 [37,38] Let $\kappa = (\mu_\kappa, \eta_\kappa)$ be a PFN, the score function $s(\kappa)$ and accuracy function $\alpha(\kappa)$ of κ are defined as follows.

$$s(\kappa) = \mu_\kappa^2 - \eta_\kappa^2 \quad (2)$$

$$\alpha(\kappa) = \mu_\kappa^2 + \eta_\kappa^2 \quad (3)$$

where $s(\kappa), \alpha(\kappa) \in [-1, 1]$.

for any two PFNs $\kappa_1 = (\mu_{\kappa_1}, \eta_{\kappa_1})$, $\kappa_2 = (\mu_{\kappa_2}, \eta_{\kappa_2})$

1. if $s(\kappa_1) > s(\kappa_2)$, then $\kappa_1 > \kappa_2$;
2. if $s(\kappa_1) < s(\kappa_2)$, then $\kappa_1 < \kappa_2$;
3. if $s(\kappa_1) = s(\kappa_2)$, then
 - (a) if $\alpha(\kappa_1) > \alpha(\kappa_2)$, then $\kappa_1 > \kappa_2$;
 - (b) if $\alpha(\kappa_1) < \alpha(\kappa_2)$, then $\kappa_1 < \kappa_2$;
 - (c) if $\alpha(\kappa_1) = \alpha(\kappa_2)$, then $\kappa_1 = \kappa_2$.

Definition 3 [37,38] For PFNs $\kappa = (\mu_\kappa, \eta_\kappa)$, $\kappa_1 = (\mu_{\kappa_1}, \eta_{\kappa_1})$ and $\kappa_2 = (\mu_{\kappa_2}, \eta_{\kappa_2})$, we have

$$\kappa_1 \oplus \kappa_2 = (\sqrt{\mu_{\kappa_1}^2 + \mu_{\kappa_2}^2 - \mu_{\kappa_1}^2 \mu_{\kappa_2}^2}, \eta_{\kappa_1} \eta_{\kappa_2}) \quad (4)$$

$$\kappa_1 \otimes \kappa_2 = (\mu_{\kappa_1} \mu_{\kappa_2}, \sqrt{\eta_{\kappa_1}^2 + \eta_{\kappa_2}^2 - \eta_{\kappa_1}^2 \eta_{\kappa_2}^2}) \quad (5)$$

$$\lambda \kappa_1 = (\sqrt{1 - (1 - \mu_{\kappa_1}^2)^\lambda}, \eta_{\kappa_1}^\lambda), \lambda > 0 \quad (6)$$

$$\kappa_1^\lambda = (\mu_{\kappa_1}^\lambda, \sqrt{1 - (1 - \eta_{\kappa_1}^2)^\lambda}), \lambda > 0 \quad (7)$$

Definition 4 [39] Let $\kappa_j = (\mu_j, \eta_j)$ ($j = 1, 2, \dots, n$) be a collection of PFNs, the PFWAM operator is defined as

PFWAM($\kappa_1, \kappa_2, \dots, \kappa_n$)

$$= \left\langle \sqrt{\sum_{t=1}^n w_t \mu_t^2}, \sqrt{1 - \sum_{t=1}^n w_t (1 - \eta_t^2)} \right\rangle \quad (8)$$

where w_t expresses the weight of PFN κ_t , $w_t \in [0, 1]$ and $\sum_{t=1}^n w_t = 1$.

2.2 Similarity Score Computation via Graph Neural Networks

Definition 5 In graph theory, a graph is a mathematical structure used to represent objects and their relationships with each other. A graph is represented as $G = (V, E)$, where V represents the set of vertices, denoting all the nodes in the

graph, and E represents the set of edges, indicating the connections between vertices.

Definition 6 [40] Top-K similarity is applied as a ground truth in SimGNN's predictions. Each expert generates a graph based on pairwise preferences. The similarity between users is used as the similarity between graph pairs. Assume there are u experts and $\#A$ alternatives. The calculation method is as follows:

$$v_i^{(u)} = \frac{\sum_{j \in A} p_{ij}^{(u)}}{\#A} \quad (9)$$

where $v_i^{(u)}$ represents the preference score of expert u for alternative i . $p_{ij}^{(u)}$ represents the pairwise preference of user u between alternatives i and j , and $\#A$ denotes the number of elements in the set of alternatives A . Each expert selects the top K alternatives based on their scores as their preferred options. Therefore, the Top-K similarity calculation method is as follows:

$$S_{i,j}^{(Top-K)} = \frac{|u_i^{(Top-K)} \cap u_j^{(Top-K)}|}{K} \quad (10)$$

where $S_{i,j}^{(Top-K)}$ represents the Top-K similarity between expert i and j , where $u_i^{(Top-K)}$ represents the set of top K preferred alternatives for expert u_i .

Definition 7 The SimGNN is a type of graph neural network designed to estimate the similarity between two graphs. By capturing node features and structural information, SimGNN effectively handles complex graph structures and performs well in tasks such as graph matching, graph classification, and other graph similarity computations. SimGNN employs Graph Convolutional Networks (GCN), attention mechanisms, and Neural Tensor Networks (NTN) to compute similarity scores between graphs. The following is a brief introduction to these three components.

(1) GCN

The main function of the GCN layer is to extract local features for each node and update the node's embedding representation using information from neighboring nodes. Its forward propagation formula is as follows:

$$H_n = \sigma \left(\sum_{m \in \mathcal{N}(n)} \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H_m W_1^{(l)} \right) \quad (11)$$

where $\mathcal{N}(n)$ represents the set of neighbors for node n . $\tilde{A}$ represents the adjacency matrix with added self-loops, symbolizing the connections between nodes. $\tilde{D}$ is the degree matrix of $\tilde{A}$, H_m is the node embedding, which represents the feature data of the node m , $W_1^{(l)}$ is the weight matrix at layer l , and σ is the activation function.

The GCN aggregates local information of nodes in the graph into their embedding vectors through layer-wise propagation, generating context-aware embedding representations for each node.

(2) Attention mechanism

To enhance the representation capability of global graph embeddings, SimGNN employs an attention mechanism that assigns different weights to each node embedding. This mechanism ensures that, when calculating the similarity of graphs, the contributions of different nodes to the final similarity score vary. The formula is as follows:

$$\mathbf{G} = \sum_{n=1}^N \sigma \left(H_n^T \tanh \left(\frac{1}{N} \sum_{m=1}^N H_m \right) W_2 \right) H_n \quad (12)$$

where $\tanh(\cdot)$ is an activation function and W_2 is a learnable weight matrix.

Finally, the graph embedding $\mathbf{G}$ is the weighted sum of node embeddings.

(3) NTN

At the core of SimGNN is the NTN, which calculates the similarity between two graphs. NTN is a binary relation model that computes the similarity score between two graphs through a tensor operation. The similarity scoring formula is as follows:

$$f(G_i, G_j) = \sigma(G_i^T W_3 G_j + v[h_i] + b) \quad (13)$$

where W_3 is a trainable three-dimensional tensor. v is the weight matrix, and b is the bias term.

According to Eqs. (9)–(13), the prediction process of the similarity score in this paper is integrated as follows:

$$f(G_i, G_j) = \text{SimGNN}(G_i, G_j, S_{i,j}^{(Top-K)}) \quad (14)$$

3 Preference-Based Graph Similarity Neural Network Multi-attribute Group Decision-Making Framework

This section describes an evaluation framework developed for the selection of electric taxis. In the first phase, on one hand, the PyFS-based linguistic scale was used to obtain experts' uncertain information to form individual decision matrices. On the other hand, we use SimGNN to predict the similarity between experts by learning from past pairwise preference experiences, thereby assigning weights to experts. The PFWAM operator is used to integrate these weights and individual decision matrices to construct the collective Pythagorean fuzzy decision matrix. In the second phase, we consider the relative importance of attributes

Table 1 PyFS linguistic scale

Linguistic terms	PyFN
Extremely high (EH)	$\kappa(0.95, 0.15)$
Very high (VH)	$\kappa(0.80, 0.25)$
High (H)	$\kappa(0.70, 0.40)$
Medium (M)	$\kappa(0.55, 0.55)$
Low (L)	$\kappa(0.45, 0.70)$
Very low (VL)	$\kappa(0.30, 0.80)$
Extremely low (EL)	$\kappa(0.20, 0.95)$

and use the CRITIC method to obtain attribute weights. In the third phase, we use the TOPSIS method to rank the alternatives. Figure 1 illustrates the flowchart of the PyFS-PPGNS-CRITIC-TOPSIS framework.

Phase 1. Construct PyFS fuzzy the collective decision matrix.

Step 1. Establish the PyFS evaluation matrix.

Let $A = \{a_1, a_2, \dots, a_m\}$ is the set of alternatives, $\varepsilon = \{e_1, e_2, \dots, e_k\}$ is the set of experts and $C = \{c_1, c_2, \dots, c_n\}$ represents the set of attributes. The $W = \{w_1, w_2, \dots, w_n\}$ denotes the weight vector of attributes, which satisfies $w_i \in [0, 1]$ and $\sum_{i=1}^n w_i = 1$. The $\varpi = \{\tau_1, \tau_2, \dots, \tau_n\}$ denotes the weight vector of experts, which satisfies $\tau_i \in [0, 1]$ and $\sum_{i=1}^n \tau_i = 1$. Both the weights of experts and the weights of attributes are completely unknown. Then, the individual evaluation matrix $M^k = (\kappa_{ij}^k)_{m \times n}$ is given by the expert according to the linguistic scale (see Table 1) and established as Eq. (15)

$$M^k = (\kappa_{ij}^k)_{m \times n} = \begin{pmatrix} (\mu_{11}^k, \eta_{11}^k) & (\mu_{12}^k, \eta_{12}^k) & \dots & (\mu_{1n}^k, \eta_{1n}^k) \\ (\mu_{21}^k, \eta_{21}^k) & (\mu_{22}^k, \eta_{22}^k) & \dots & (\mu_{2n}^k, \eta_{2n}^k) \\ \vdots & \vdots & \ddots & \vdots \\ (\mu_{m1}^k, \eta_{m1}^k) & (\mu_{m2}^k, \eta_{m2}^k) & \dots & (\mu_{mn}^k, \eta_{mn}^k) \end{pmatrix} \quad (15)$$

Step 2. Obtain the pairwise preference matrix for each expert.

If expert e_k expresses a preference for alternative a_i over alternative a_j , which satisfies $a_i, a_j \in A$, then the corresponding pairwise preference $P_{(a_i, a_j)}$ is 1. If the preferences are the same, $P_{(a_i, a_j)}$ is 0.5. If the preference for alternative a_i is less than for alternative a_j , then $P_{(a_i, a_j)}$ is 0. Using this method, the individual pairwise preference matrix is represented as Eq. (16)

$$P^k = (p_{ij}^k)_{m \times m} = \begin{pmatrix} 1 & p_{12}^k & \dots & p_{1m}^k \\ p_{21}^k & 1 & \dots & p_{2m}^k \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1}^k & p_{m2}^k & \dots & 1 \end{pmatrix} \quad (16)$$

Step 3. Use SimGNN to predict similarity and obtain expert weights.

Convert the pairwise preferences into a graph form, as follows Eq. (17).

$$G_k = \text{Graph}(P^k) \quad (17)$$

Use Eqs. (9) and (10) to obtain the Top-K similarity, and apply it in the model for expert similarity score prediction, as follows Eq. (18).

$$f(G_k, G_{k'}) = \text{SimGNN}(G_k, G_{k'}, S_{k, k'}^{(Top-K)}) \quad (18)$$

where $f(G_k, G_{k'})$ is the predicted similarity score between experts k and k' .

Normalize the similarity scores using Eq. (19).

$$f^i(G_k, G_{k'}) = \frac{f(G_k, G_{k'}) - f_{\min}(G_k, G_{k'})}{f_{\max}(G_k, G_{k'}) - f_{\min}(G_k, G_{k'})} \quad (19)$$

Then, assign a weight to each expert using Eq. (20).

$$\tau_k = \frac{\sum_{k'=1}^n f^i(G_k, G_{k'})}{\sum_{k=1}^n \sum_{k'=1}^n f^i(G_k, G_{k'})} \quad (20)$$

where $\tau_k \in [0, 1]$.

Step 4. Construct the collective Pythagorean fuzzy decision matrix T_{ij} .

From Eq. (15), the individual evaluation matrices of k experts can be obtained. Use the PFWAM operator given by Eq. (8) to aggregate the individual evaluation matrix M^k and the weights of the experts τ_k , obtaining the collective Pythagorean fuzzy decision matrix T_{ij} , as follows Eq. (21).

$$T_{ij} = \text{PFWAM}(M^1, M^2, \dots, M^k) = \begin{pmatrix} (\mu_{11}, \eta_{11}) & (\mu_{12}, \eta_{12}) & \dots & (\mu_{1n}, \eta_{1n}) \\ (\mu_{21}, \eta_{21}) & (\mu_{22}, \eta_{22}) & \dots & (\mu_{2n}, \eta_{2n}) \\ \vdots & \vdots & \ddots & \vdots \\ (\mu_{m1}, \eta_{m1}) & (\mu_{m2}, \eta_{m2}) & \dots & (\mu_{mn}, \eta_{mn}) \end{pmatrix} \quad (21)$$

Phase 2. Determine the attribute weights.

Step 5. Compute the score function matrix $Y = (y_{ij})_{m \times n}$.

According to Eq. (2), obtain the score values of the evaluation matrix Y in the following way Eq. (22):

$$y_{ij} = s(\kappa) = \mu_{\kappa}^2 - \eta_{\kappa}^2 \quad (22)$$

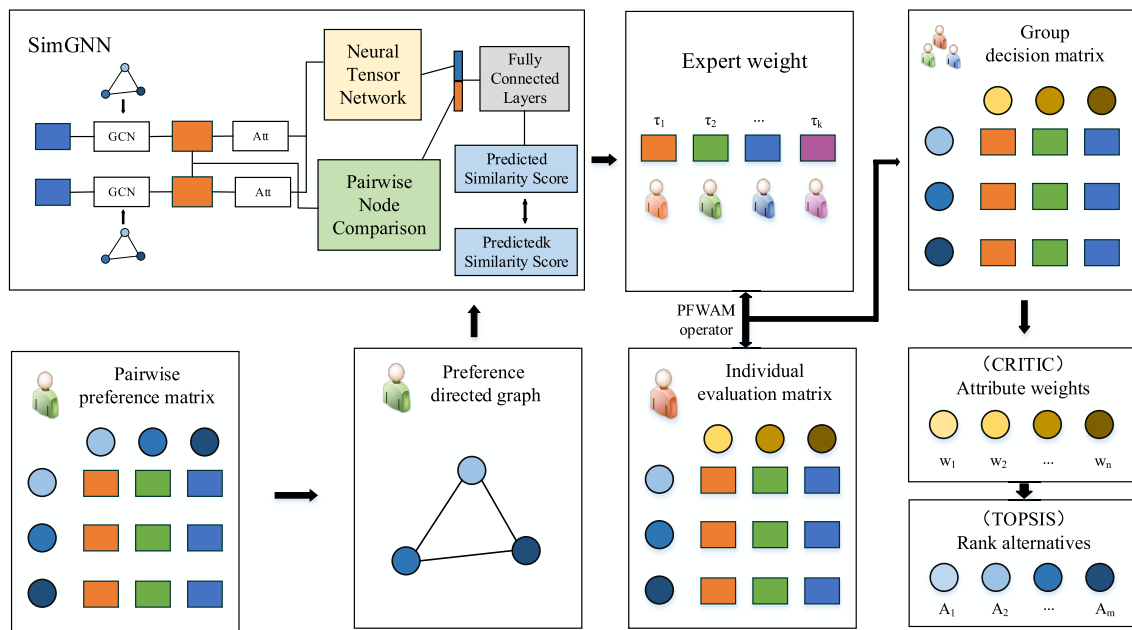


Fig. 1 PyFS-PPGNS-CRITIC-TOPSIS framework

Step 6. Convert matrix Y into an orthogonal Pythagorean fuzzy matrix $Y^\lambda = (y_{ij}^\lambda)_{m \times n}$ by Eq. (23).

$$y_{ij}^\lambda = \begin{cases} \frac{y_{ij} - y_j^-}{y_j^+ - y_j^-} & \text{if } j \in B \\ \frac{y_j^+ - y_{ij}}{y_j^+ - y_j^-} & \text{if } j \in C \end{cases} \quad (23)$$

where B and C represents benefit attribute and cost attribute. The $y_j^+ = \max_i \{y_{ij}\}$ and $y_j^- = \min_i \{y_{ij}\}$.

Step 7. Calculate the standard deviation by Eq. (24).

$$\sigma_j = \sqrt{\frac{\sum_{i=1}^m (y_{ij}^\lambda - \bar{y}_j)^2}{m}} \quad (24)$$

where $\bar{y}_j = \frac{\sum_{i=1}^m y_{ij}^\lambda}{m}$. The m is the number of alternatives.

Step 8. Calculate the correlation between alternative pairs by Eq. (25).

$$\rho_{jf} = \frac{\sum_{i=1}^m (y_{ij}^\lambda - \bar{y}_j)(y_{if}^\lambda - \bar{y}_f)}{\sqrt{\sum_{i=1}^m (y_{ij}^\lambda - \bar{y}_j)^2 \sum_{i=1}^m (y_{if}^\lambda - \bar{y}_f)^2}} \quad (25)$$

Step 9. Calculate the quantity of information of the attributes by Eq. (26).

$$z_j = \sigma_j \sum_{f=1}^n (1 - \rho_{jf}) \quad (26)$$

Step 10. Calculate the objective weights of the attributes by Eq. (27).

$$w_j = \frac{z_j}{\sum_{j=1}^n z_j} \quad (27)$$

Phase 3. Rank the alternatives.

Step 11. Calculate the positive ideal solutions (PIS) and negative ideal solutions (NIS) by Eqs. (28) and (29).

$$\begin{aligned} a^+ &= \{C_j, \max_i \{s(C_j(a_i))\} | j = 1, 2, \dots, n\} \\ &= \{ \langle (C_1, \kappa(\mu_1^+, \eta_1^+)) \rangle, \langle (C_2, \kappa(\mu_2^+, \eta_2^+)) \rangle, \dots, \\ &\quad \langle (C_n, \kappa(\mu_n^+, \eta_n^+)) \rangle \} \end{aligned} \quad (28)$$

$$\begin{aligned} a^- &= \{C_j, \min_i \{s(C_j(a_i))\} | j = 1, 2, \dots, n\} \\ &= \{ \langle (C_1, \kappa(\mu_1^-, \eta_1^-)) \rangle, \langle (C_2, \kappa(\mu_2^-, \eta_2^-)) \rangle, \dots, \\ &\quad \langle (C_n, \kappa(\mu_n^-, \eta_n^-)) \rangle \} \end{aligned} \quad (29)$$

Step 12. Calculate the distance of each alternative to the PIS and NIS.

In this paper, the Euclidean distance is used to calculate the distances of the alternatives, as follows Eqs. (30) and (31).

$$\begin{aligned} D(a_i, a^+) &= \sqrt{\frac{1}{2} \sum_{j=1}^n w_j \left(|(\mu_{ij})^2 - (\mu^+)^2| + |(\eta_{ij})^2 - (\eta^+)^2| \right)} \end{aligned} \quad (30)$$

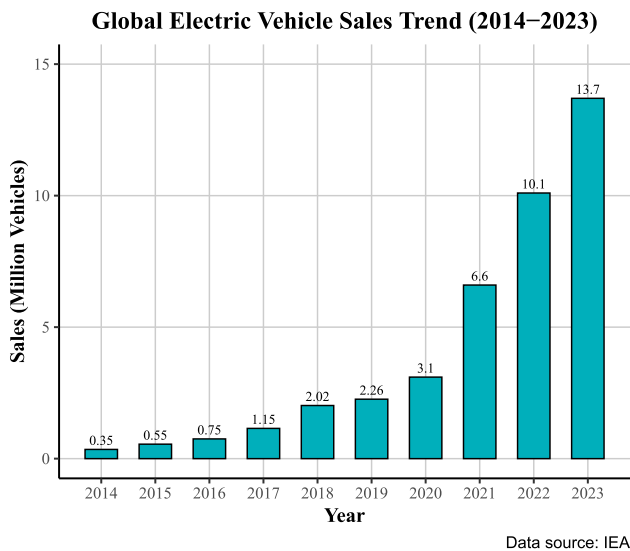


Fig. 2 Global electric vehicle sales over the past decade

$$D(a_i, a^-) = \sqrt{\frac{1}{2} \sum_{j=1}^n w_j \left(|(\mu_{ij})^2 - (\mu^-)^2| + |(\eta_{ij})^2 - (\eta^-)^2| \right)} \quad (31)$$

Step 13. Calculate the index of relative closeness of alternatives by Eq. (32)

$$\varphi(a_i) = \frac{D(a_i, a^-)}{D_{\max}(a_i, a^-)} - \frac{D(a_i, a^+)}{D_{\min}(a_i, a^+)}; i = 1, 2, \dots, n \quad (32)$$

where $\varphi(a_i)$ satisfies $\varphi(a_i) \leq 0$ and the larger the value of $\varphi(a_i)$, the better the alternative a_i .

Step 14. Determine the best alternative.

Rank the values of $\varphi(a_i)$ from highest to lowest; the alternative with the highest rank is selected as the best alternative.

Table 2 Standard evaluation system

ID	Attribute	Unit	Definition	References
c ₁	Acceleration time (0–100 km/h)	s	Time to accelerate from 0 to 100 km/h	[41]
c ₂	Top speed	km/h	Maximum speed achieved by the vehicle	[42]
c ₃	Battery capacity	kw	Rated capacity of the battery	[43]
c ₄	Charge time	min	Charging time to full capacity	[7]
c ₅	Distance traveled	km	Driving range on a full battery charge	[44]

4 Numerical Examples

This section introduces the application of the proposed method to solve vehicle selection problem under complex uncertainties, demonstrating the effectiveness and practicality of the proposed method.

Today, the world is facing unprecedented climate and energy challenges, and many countries are addressing these issues by developing NEVs. Figure 2 displays the global electric vehicle sales over the past decade. In 2024, the global electric vehicle market continues its rapid growth trend. According to data from the China Automobile Dealers Association (CADA), global vehicle sales reached 58.24 million units from January to August 2024, with 9.39 million being NEVs, accounting for 16.1%. Therefore, it is expected that the global electric vehicle market will continue to expand in the coming years, with 30% of all vehicles projected to be electric by 2030 [41]. Consequently, it is essential to establish an appropriate decision-making framework for selecting electric vehicles.

We selected the 10 most commonly used NEV models on the current market as evaluation alternatives ($a_1, a_2, a_3, \dots, a_{10}$), with prices remaining generally consistent and within the range suitable for vehicle selection. Additionally, based on references and surveys, we set 5 evaluation dimensions, namely, acceleration time (0–100 km/h) (C_1) [41], top speed (C_2) [42], battery capacity (C_3) [43], charge time (C_4) [7], distance traveled (C_5) [44], as detailed in Table 2. Finally, we formed a panel of 5 experts to evaluate the importance of different cars under various criteria according to Table 2, and then provide the pairwise preference matrices for different cars. These experts come from various fields related to NEVs and have over 10 years of driving experience. Details are provided in Table 3. Prior to this, we had trained and fine-tuned the parameters of the SimGNN model for obtaining expert weights. The specific implementation process is as follows:

Phase I. Construct PyFS fuzzy the collective decision matrix.

Step 1. Table 4 provides the evaluation information for the 10 alternatives across the 5 evaluation dimensions.

Step 2. The pairwise preference matrix for each expert is as follows:

Table 3 Expert information

Experts	Field	Driving experience	Age
e_1	Transportation sector	15	45
e_2	Taxi driver	21	48
e_3	Taxi driver	13	37
e_4	Passenger	18	40
e_5	Transportation sector	20	51

$$\begin{aligned}
 P^1 &= \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix} \\
 P^2 &= \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \\
 P^3 &= \begin{pmatrix} 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \end{pmatrix} \\
 P^4 &= \begin{pmatrix} 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \\
 P^5 &= \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}
 \end{aligned}$$

Step 3. By use Eq. (20), the similarity score between experts is obtained as follows:

	e_1	e_2	e_3	e_4	e_5
e_1	1	0.8596	0.8759	0.8952	0.9696
e_2	0.8310	1	0.8795	0.8474	0.8623
e_3	0.8259	0.8392	1	0.8573	0.8023
e_4	0.8851	0.8399	0.8153	1	0.9036
e_5	0.8315	0.8597	0.7808	0.8898	1

The difference in similarity scores between the same pair of experts arises because the neural network employs an attention mechanism to select important nodes within the graph, which may result in varying similarity scores for the graph pairs.

Subsequently, the weights of the experts can be obtained using Eqs. (21) and (22) as follows:

$$\varpi_1 = 0.30, \varpi_2 = 0.20, \varpi_3 = 0.12, \varpi_4 = 0.22, \varpi_5 = 0.16$$

Step 4. Use the PFWAM operator defined in Eq. (8) to obtain the collective Pythagorean fuzzy decision matrix listed in Table 5.

Phase II. Determine the attribute weights.

Step 5. Use Eq. (20) to obtain the score function matrix Y shown in Table 6.

Step 6. Convert matrix Y into an orthogonal Pythagorean fuzzy matrix Y^1 shown in Table 7.

Step 7. Calculate the standard deviation use Eq. (22), the results are as follows:

$$\sigma_1 = 0.3328, \sigma_2 = 0.2190, \sigma_3 = 0.1854, \sigma_4 = 0.1478, \sigma_5 = 0.2043$$

Step 8. Use Eq. (23) to calculate the correlation between the attributes, as shown in Table 8.

Step 9. Use Eq. (24) to calculate the quantity of information of the attributes, the results are as follows:

$$z_1 = 1.4278, z_2 = 0.9176, z_3 = 0.9548, z_4 = 0.5128, z_5 = 0.8662$$

Step 10. Finally, use Eq. (25) to obtain the objective weights of the attributes, the results are as follows:

$$w_1 = 0.3051, w_2 = 0.1961, w_3 = 0.2041, w_4 = 0.1096, w_5 = 0.1851$$

Phase III. Rank the alternatives.

Step 11. Use Eqs. (26) and (27) to calculate the PIS and NIS, the results are as follows:

$$a^+ = \{\kappa(0.8577, 0.5142), \kappa(0.8258, 0.5639), \kappa(0.7964, 0.6048), \kappa(0.7535, 0.6575), \kappa(0.8046, 0.5938)\}$$

$$a^- = \{\kappa(0.5261, 0.8505), \kappa(0.5624, 0.8268), \kappa(0.5784, 0.8158), \kappa(0.5869, 0.8097), \kappa(0.5193, 0.8546)\}$$

Step 12. Calculate the weighted distance between the car alternatives and the PIS and NIS by Eqs. (28) and Eq. (29), the results are shown in Table 9.

Step 13. Calculate the relative closeness index of the alternatives using Eq. (30), as shown in Table 9.

Step 14. Finally, the ranking order of the alternatives based on the values of $\varphi(a_i)$ is as follows: $a_9 \succ a_5 \succ a_1 \succ a_7 \succ a_2 \succ a_6 \succ a_{10} \succ a_3 \succ a_8 \succ a_4$. Therefore, the best vehicle for NEVs is a_9 .

5 Discussions

This section primarily performs sensitivity and comparative analysis of the proposed PyFS-PPGNS-CRITIC-TOPSIS

Table 4 Experts evaluation information

	e_1					e_2					e_3					e_4					e_5				
	c_1	c_2	c_3	c_4	c_5	c_1	c_2	c_3	c_4	c_5	c_1	c_2	c_3	c_4	c_5	c_1	c_2	c_3	c_4	c_5	c_1	c_2	c_3	c_4	c_5
a_1	VH	H	M	L	M	M	VL	EH	VH	H	EH	EH	EH	M	M	M	M	H	EH	VH	EH	M	EH	M	H
a_2	EH	EH	L	M	H	H	M	EH	EL	M	H	H	M	VH	VH	H	VL	M	EH	M	M	H	VH	M	L
a_3	H	M	L	VH	L	VL	VH	EH	M	EH	M	M	H	VL	L	VL	EH	H	M	H	H	EH	M	M	M
a_4	M	H	M	VL	M	M	VL	H	VH	H	EH	M	EH	H	M	M	M	M	VH	VH	VL	M	VH	EH	VH
a_5	EH	VH	L	M	H	VH	M	EH	H	L	H	EH	M	M	VH	EH	H	L	EH	M	H	L	EH	H	M
a_6	H	VL	H	L	VH	VL	EH	M	H	VH	M	EH	L	VH	H	H	EH	VH	M	VL	EH	H	H	VH	L
a_7	EH	M	M	M	H	M	H	VH	EH	EH	EH	M	H	VL	VH	M	M	H	L	VH	M	EH	L	M	VH
a_8	M	VH	L	VH	L	H	M	EH	M	L	M	H	EH	M	L	VL	H	VH	EH	M	L	M	EH	VL	H
a_9	H	EH	M	M	VH	EH	VL	H	M	H	H	EH	L	M	H	EH	VH	L	H	H	EH	EH	H	M	M
a_{10}	EH	H	L	L	M	L	EH	VH	VH	M	EH	M	L	VH	M	L	M	H	EH	M	M	L	EH	VH	VH

Table 5 The collective Pythagorean fuzzy decision matrix T

	c_1	c_2	c_3	c_4	c_5
a_1	(0.7552, 0.6555)	(0.6241, 0.7813)	(0.7941, 0.6078)	(0.6884, 0.7253)	(0.6667, 0.7453)
a_2	(0.7644, 0.6448)	(0.6977, 0.7164)	(0.6668, 0.7452)	(0.6523, 0.7579)	(0.6199, 0.7847)
a_3	(0.5459, 0.8378)	(0.7729, 0.6345)	(0.6759, 0.7370)	(0.6145, 0.7889)	(0.6494, 0.7604)
a_4	(0.5853, 0.8108)	(0.5624, 0.8268)	(0.6824, 0.7310)	(0.7056, 0.7086)	(0.6840, 0.7294)
a_5	(0.8577, 0.5142)	(0.7093, 0.7049)	(0.6806, 0.7327)	(0.7094, 0.7048)	(0.6164, 0.7875)
a_6	(0.6722, 0.7404)	(0.7713, 0.6365)	(0.6717, 0.7409)	(0.6353, 0.7722)	(0.6561, 0.7546)
a_7	(0.7452, 0.6668)	(0.6583, 0.7527)	(0.6472, 0.7623)	(0.6113, 0.7914)	(0.8046, 0.5938)
a_8	(0.5261, 0.8505)	(0.6844, 0.7291)	(0.7964, 0.6048)	(0.7103, 0.7039)	(0.5193, 0.8546)
a_9	(0.8536, 0.5209)	(0.8258, 0.5639)	(0.5784, 0.8158)	(0.5869, 0.8097)	(0.7111, 0.7031)
a_{10}	(0.7163, 0.6978)	(0.6801, 0.7331)	(0.6805, 0.7328)	(0.7535, 0.6575)	(0.5959, 0.8030)

evaluation framework to demonstrate the effectiveness of the method.

5.1 Sensitivity Analysis

Sensitivity analysis aims to examine how changes in different conditions during the decision-making process affect the decision outcomes. The number of iterations in SimGNN training affects expert weight prediction and, consequently, the ranking of car alternatives. Therefore, we analyzed the following two aspects:

(1) The impact of iteration parameters on expert weights

We set four iteration parameters: 50, 100, 150 and 200. Figure 3 shows the changes in expert weights with different iteration counts. From Fig. 3, we can derive valuable insights. The weight of expert e_3 increases steadily with the number of epochs, indicating that e_3 's influence within the group gradually strengthens over time. In contrast, the weights of experts e_1 and e_5 decrease consistently, suggesting that their opinions become progressively less significant. Meanwhile, the weights of experts e_2 and e_4 show considerable fluctua-

Table 6 Score function matrix

	c_1	c_2	c_3	c_4	c_5
a_1	0.1406	-0.2209	0.2612	-0.0522	-0.1110
a_2	0.1685	-0.0264	-0.1107	-0.1489	-0.2315
a_3	-0.4039	0.1948	-0.0863	-0.2448	-0.1565
a_4	-0.3148	-0.3673	-0.0687	-0.0042	-0.0642
a_5	0.4712	0.0062	-0.0736	0.0065	-0.2402
a_6	-0.0963	0.1898	-0.0978	-0.1927	-0.1390
a_7	0.1107	-0.1332	-0.1622	-0.2526	0.2948
a_8	-0.4466	-0.0632	0.2685	0.0091	-0.4607
a_9	0.4573	0.3640	-0.3310	-0.3112	0.0113
a_{10}	0.0262	-0.0749	-0.0739	0.1355	-0.2897

tions, indicating that their influence varies at different stages. Notably, when the epoch reaches 150, both e_2 and e_4 experience a sharp change, suggesting a divergence from the group at that point.

(2) The impact of iteration parameters on the ranking of alternatives

Table 7 Orthogonal Pythagorean fuzzy matrix

	c_1	c_2	c_3	c_4	c_5
a_1	0.3548	0.2573	0.7746	0.5617	0.3752
a_2	0.3248	0.4659	0.3755	0.6655	0.2459
a_3	0.9391	0.7033	0.4017	0.7683	0.3264
a_4	0.8435	0.1002	0.4206	0.5102	0.4255
a_5	0.0000	0.5010	0.4153	0.4987	0.2366
a_6	0.6091	0.6980	0.3894	0.7124	0.3452
a_7	0.3869	0.3514	0.3202	0.7768	0.8106
a_8	0.9849	0.4265	0.7824	0.4960	0.0000
a_9	0.0150	0.8849	0.1392	0.8396	0.5065
a_{10}	0.4776	0.4140	0.4150	0.3603	0.1834

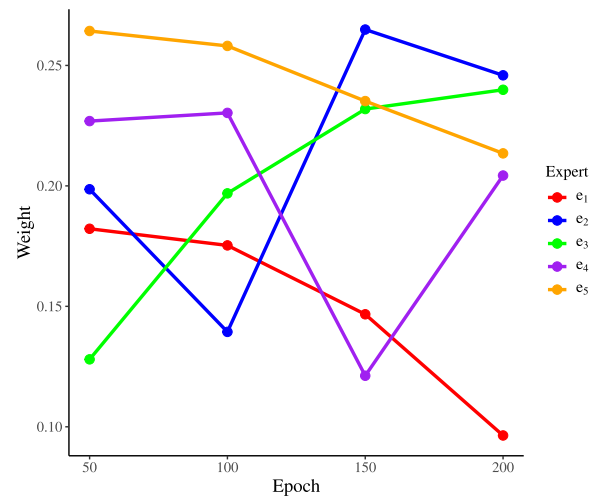
Table 8 Calculate the correlation between the attributes

	c_1	c_2	c_3	c_4	c_5
c_1	1.00	-0.26	0.44	-0.16	-0.31
c_2	-0.26	1.00	-0.52	0.61	-0.02
c_3	0.44	-0.52	1.00	-0.54	-0.53
c_4	-0.16	0.61	-0.54	1.00	0.62
c_5	-0.31	-0.02	-0.53	0.62	1.00

Table 9 Weighted distance results and relative closeness index

	$D(a_i, a^+)$	$D(a_i, a^-)$	$\varphi(a_i)$	Ranking
a_1	0.3954	0.4593	-0.2585	3
a_2	0.4329	0.4241	-0.4395	5
a_3	0.4989	0.3441	-0.7939	8
a_4	0.5166	0.3168	-0.9005	10
a_5	0.3558	0.4906	-0.0787	2
a_6	0.4463	0.4100	-0.5069	6
a_7	0.4111	0.4453	-0.3325	4
a_8	0.5087	0.3295	-0.8518	9
a_9	0.3377	0.5033	0.0000	1
a_{10}	0.4472	0.4090	-0.5116	7

The impact of the number of iterations on the ranking of the alternatives is shown in Table 10. Clearly, under the four different parameter settings, the rankings of alternatives a_2, a_3, a_6 and a_8 show slight fluctuations. However, alternative a_9 is consistently ranked as the best choice, and alternative a_4 is consistently ranked as the worst. Therefore, the developed method proves to be robust against changes in decision rankings. Figure 4 provides a clearer view of the changes in alternative rankings.

**Fig. 3** Weight changes of experts over epochs

5.2 Comparative Analysis

To highlight the advantages and contributions of the proposed method, it was compared with the recent MAGDM method, as shown in Table 12. The comparison focuses on two aspects: the determination of weights and the method for ranking alternatives.

(1) The determination of weights

The determination of weights involves two key components: expert weights and criterion weights. Peng et al. [45] subjectively assigned expert weights based on experience and developed a combined subjective-objective approach for criterion weights, utilizing the CRITIC method for objective weights. Rani et al. [47] proposed a new similarity measure between picture fuzzy sets to objectively determine the criterion weights, but unfortunately, the expert weights were also subjectively assigned, which led to significant uncertainty. Sarkar and Biswas [46] proposed a distance measure method based on Pythagorean fuzzy set cross-evaluation. This method utilizes both subjective and objective criterion weights, modifying AHP with Pythagorean fuzzy information to calculate subjective weights, and employing a Pythagorean fuzzy entropy weight model to determine objective weights. Wang et al. [18] introduced a weighted distance measure based on PyFS to derive the importance of experts. The CRITIC method was used to determine the attribute weights, which helps provide more objective decision-making results. Compared to the methods mentioned above, our proposed expert weight calculation method, based on pairwise preference graph neural similarity, offers the advantage of incorporating two types of data and deeply analyzing expert evaluation information. Additionally, as it leverages neural networks, it can adaptively learn from and manage large-scale expert data, minimizing human intervention and

Table 10 Sensitivity analysis for training epochs and decision results

	Epoch-50		Epoch-100		Epoch-150		Epoch-200	
	$\varphi(a_i)$	Ranking	$\varphi(a_i)$	Ranking	$\varphi(a_i)$	Ranking	$\varphi(a_i)$	Ranking
a_1	-0.1060	3	-0.2585	3	-0.1936	3	-0.1107	3
a_2	-0.3411	5	-0.4395	5	-0.3807	5	-0.3365	6
a_3	-0.6073	9	-0.7939	8	-0.6291	8	-0.5572	9
a_4	-0.7273	10	-0.9005	10	-0.7780	10	-0.6985	10
a_5	-0.0201	2	-0.0787	2	-0.0533	2	-0.0281	2
a_6	-0.3680	6	-0.5069	6	-0.3907	6	-0.3364	5
a_7	-0.2600	4	-0.3325	4	-0.3090	4	-0.2585	4
a_8	-0.5869	8	-0.8518	9	-0.6474	9	-0.5414	8
a_9	0.0000	1	0.0000	1	0.0000	1	0.0000	1
a_{10}	-0.3722	7	-0.5116	7	-0.3943	7	-0.3456	7

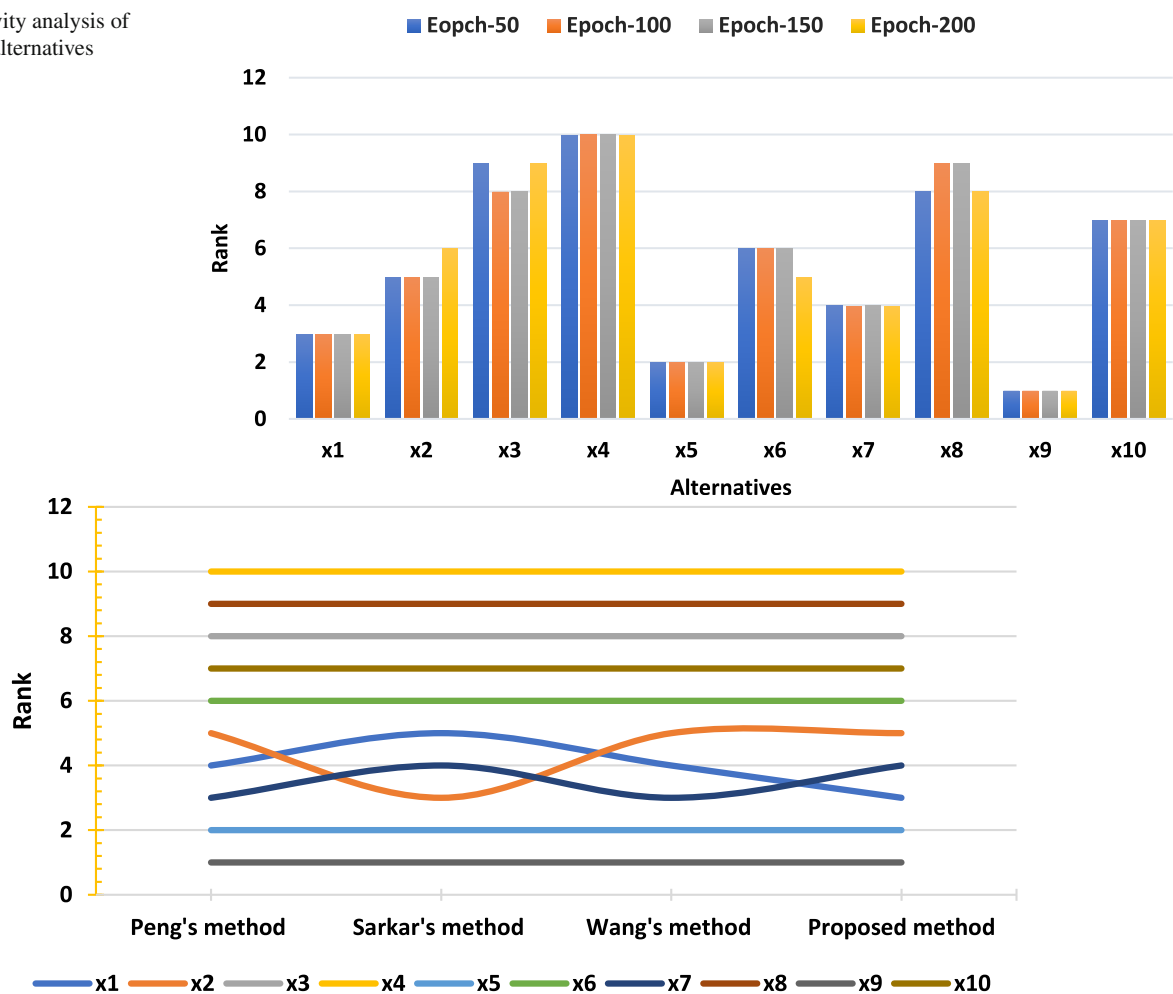
Fig. 4 Sensitivity analysis of the ranking of alternatives**Fig. 5** Ranking results of four methods

Table 11 Decision results of different methods

Methods	Ranking	Best alternative
Peng's method [45]	$a_9 > a_5 > a_7 > a_1 > a_2 > a_6 > a_{10} > a_3 > a_8 > a_4$	a_9
Sarkar's method [46]	$a_9 > a_5 > a_2 > a_7 > a_1 > a_6 > a_{10} > a_3 > a_8 > a_4$	a_9
Wang's method [18]	$a_9 > a_5 > a_7 > a_1 > a_2 > a_6 > a_{10} > a_3 > a_8 > a_4$	a_9
Proposed method	$a_9 > a_5 > a_1 > a_7 > a_2 > a_6 > a_{10} > a_3 > a_8 > a_4$	a_9

Table 12 Comparison with existing MAGDM methods

References	Information representation	Determination of expert weights	Determination of attribute weights	Ranking method
Proposed method	PyFS	Objective	CRITIC	TOPSIS
Peng's method [45]	PyFS	Subjective	CRITIC	CoCoSo
Rani's method [47]	PFS	Subjective	Based on a new similarity measure	MARCOS
Sarkar's method [46]	PyFS	–	AHP and Entropy weight method	TOPSIS
Wang's method [18]	PyFS	Objective	CRITIC	MARCOS

In what follows, we provide the meanings of some abbreviations: *AHP* analytic hierarchy process, *CoCoSo* combined compromise solution, *MARCOS* measurement of alternatives and ranking according to compromise solution, *TOPSIS* technique for order preference by similarity to ideal solution

bias, thus improving the objectivity and fairness of weight distribution.

(2) The method for ranking alternatives

To more objectively validate the effectiveness of our proposed decision framework, we compared it with three existing MCGDM methods. These include the PyFS-CRITIC-CoCoSo method [45], the PyFS-AHP-TOPSIS method [46], and the PyFS-CRITIC-MARCOS method [18]. Using the available evaluation data, we assessed 10 car alternatives. The specific results are shown in Table 11. The specific changes are shown in Fig. 5. Comparative analysis reveals that our proposed method shows high consistency with the rankings provided by other group decision methods, all indicating that alternative x_9 is the best choice. However, due to different focuses of each method, the rankings of some alternatives vary slightly. The possible reasons for difference can be seen as follows.

1. Different methods of weight determination

Our expert weight determination method incorporates two types of information: experts' pairwise preferences between alternatives and their evaluations of alternatives under different attributes. By transforming this information into a graph structure, we leverage neural networks to learn and determine expert weights. In contrast, Peng et al. [45] based expert weights solely on experience, which introduces clear risks. Although Wang et al. [18] introduced a weighted metric method to infer expert weights, their approach did not account for relationships between experts. Therefore, our proposed method offers a more comprehensive solution.

2. Different methods for ranking alternatives.

Table 12 presents the ranking methods employed by scholars. The TOPSIS method emphasizes the distance to ideal and negative ideal solutions, ranking alternatives based on their relative closeness. The CoCoSo method combines weighted sums and geometric means to determine compromise solutions, incorporating more complex calculations. The MARCOS method calculates utility values based on positive and negative ideal solutions as well as reference solutions, focusing on decision-making through utility values, which offers greater flexibility. This study selects the CRITIC-TOPSIS method because it provides a more representative ranking of alternatives by comprehensively considering objective weights and distance evaluations. Additionally, the decision-making process of the TOPSIS method is relatively straightforward, whereas the complexities of the CoCoSo [45] and MARCOS [18] methods may compromise the accuracy of the ranking results.

6 Conclusions

In the context of global carbon reduction, evaluating and selecting NEV has significant practical value. In this paper, we establish a comprehensive decision-making framework, PyFS-PPGNS-CRITIC-TOPSIS, to address vehicle selection problems. First, we collected two types of information: the first type is experts' pairwise preferences between alternatives, and the second type is experts' evaluation information for alternatives under different attributes. Importantly, pairwise preferences and evaluation information serve different functions. We converted the pairwise preference information

into graph format and used a graph similarity neural network to predict similarity scores between these graphs, reflecting the similarity between experts. A higher similarity score indicates greater consensus between the expert and others, leading to a higher assigned weight, and vice versa. Next, we introduced the PFWAM operator to integrate individual evaluation information with expert weights, resulting in the group decision matrix. Considering conflicts between attributes, we calculated attribute weights using the PyFS-CRITIC method and ranked the alternatives using the PyFS-TOPSIS method. Finally, we validated the effectiveness of the proposed framework through numerical examples, sensitivity analysis, and comparisons with other current group decision-making methods.

The advantages of the method proposed in this paper are as follows: (1) Faster and more objective determination of expert weights based on historical data; (2) Better exploration of expert relationships using graph structures; (3) More efficient decision-making results through the use of the CRITIC-TOPSIS method. Although the proposed framework demonstrates good applicability for solving vehicle selection problems, there is still room for improvement. First, since the PPGNS method involves neural networks, further research is needed to enhance its interpretability. Additionally, as it involves vehicle selection, incorporating performance data of the vehicles into the decision-making process could help reduce uncertainty in expert judgments. Finally, while the framework is set within a Pythagorean fuzzy environment, other fuzzy environments, such as spherical fuzzy sets, picture fuzzy sets, and q-ROF fuzzy sets, also have their advantages. Future research could explore these alternatives.

Data availability This manuscript does not contain any data beyond what is within the article itself.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

References

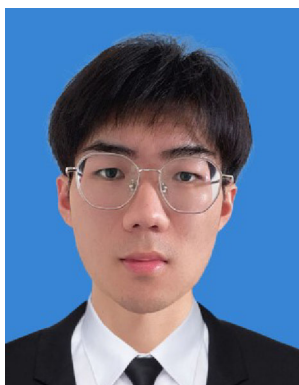
1. Zhou, W., Yang, L., Cai, Y., Ying, T.: Dynamic programming for new energy vehicles based on their work modes part II: fuel cell electric vehicles. *J. Power Sources* **407**, 92–104 (2018). <https://doi.org/10.1016/j.jpowsour.2018.10.048>
2. Hu, L., Dong, J., Lin, Z., Yang, J.: Analyzing battery electric vehicle feasibility from taxi travel patterns: the case study of New York city, USA. *Transp. Res. Part C Emerg. Technol.* **87**, 91–104 (2018). <https://doi.org/10.1016/j.trc.2017.12.017>
3. Meckling, J., Nahm, J.: The politics of technology bans: industrial policy competition and green goals for the auto industry. *Energy Policy* **126**, 470–479 (2019). <https://doi.org/10.1016/j.enpol.2018.11.031>
4. Ecer, F.: A consolidated MCDM framework for performance assessment of battery electric vehicles based on ranking strategies. *Renew. Sustain. Energy Rev.* **143**, 110916 (2021). <https://doi.org/10.1016/j.rser.2021.110916>
5. Golui, S., Mahapatra, B.S., Mahapatra, G.S.: A new correlation-based measure on Fermatean fuzzy applied on multi-criteria decision making for electric vehicle selection. *Expert Syst. Appl.* **237**, 121605 (2024). <https://doi.org/10.1016/j.eswa.2023.121605>
6. Tian, Zp., Liang, Hm., Nie, rx., Wang, Xk., Wang, Jq.: Data-driven multi-criteria decision support method for electric vehicle selection. *Comput. Ind. Eng.* **177**, 109061 (2023). <https://doi.org/10.1016/j.cie.2023.109061>
7. Sonar, H.C., Kulkarni, S.D.: An integrated AHP-MABAC approach for electric vehicle selection. *Res. Transp. Bus. Manag.* **41**, 100665 (2021). <https://doi.org/10.1016/j.rtbm.2021.100665>
8. Wan, S.-P., Wu, H., Dong, J.-Y.: An integrated method for complex heterogeneous multi-attribute group decision-making and application to photovoltaic power station site selection. *Expert Syst. Appl.* **242**, 122456 (2024). <https://doi.org/10.1016/j.eswa.2023.122456>
9. Mishra, A.R., Rani, P., Alrasheedi, A.F., Dwivedi, R.: Evaluating the blockchain-based healthcare supply chain using interval-valued Pythagorean fuzzy entropy-based decision support system. *Eng. Appl. Artif. Intell.* **126**, 107112 (2023). <https://doi.org/10.1016/j.engappai.2023.107112>
10. Wu, P., Zhou, L., Martínez, L.: An integrated hesitant fuzzy linguistic model for multiple attribute group decision-making for health management center selection. *Comput. Ind. Eng.* **171**, 108404 (2022). <https://doi.org/10.1016/j.cie.2022.108404>
11. Gou, X., Deng, F., Xu, X., Zhou, W., Herrera-Viedma, E.: Medical health resources allocation evaluation in public health emergencies by an improved Oreste method with linguistic preference orderings. *Fuzzy Optim. Decis. Mak.* **23**(1), 1–27 (2024). <https://doi.org/10.1007/s10700-023-09409-3>
12. Zhang, R., Xu, Z., Gou, X.: Electre II method based on the cosine similarity to evaluate the performance of financial logistics enterprises under double hierarchy hesitant fuzzy linguistic environment. *Fuzzy Optim. Decis. Mak.* **22**(1), 23–49 (2023). <https://doi.org/10.1007/s10700-022-09382-3>
13. Zadeh, L.A.: Fuzzy sets. *Inf. Control* **8**, 338–353 (1965). [https://doi.org/10.1016/S0019-9958\(65\)90241-X](https://doi.org/10.1016/S0019-9958(65)90241-X)
14. Atanassov, K.T.: Intuitionistic fuzzy sets. *Fuzzy Sets Syst.* **20**, 87–96 (1986). [https://doi.org/10.1016/S0165-0114\(86\)80034-3](https://doi.org/10.1016/S0165-0114(86)80034-3)
15. Yager, R.R.: Pythagorean fuzzy subsets. In: 2013 Joint IFSA World Congress and NAFIPS Annual Meeting (IFSA/NAFIPS), pp. 57–61 (2013). <https://doi.org/10.1109/IFSA-NAFIPS.2013.6608375>
16. Lin, S.-S., Zhou, A., Shen, S.-L.: Safety assessment of excavation system via TOPSIS-based MCDM modelling in fuzzy environment. *Appl. Soft Comput.* **138**, 110206 (2023). <https://doi.org/10.1016/j.asoc.2023.110206>
17. Krishankumar, R., Dhruva, S., Ravichandran, K.S., Kar, S.: Selection of a viable blockchain service provider for data management within the internet of medical things: an MCDM approach to indian healthcare. *Inf. Sci.* **657**, 119890 (2024). <https://doi.org/10.1016/j.ins.2023.119890>
18. Wang, Y., Wang, W., Wang, Z., Deveci, M., Roy, S.K., Kadry, S.: Selection of sustainable food suppliers using the Pythagorean fuzzy CRITIC-MARCOS method. *Inf. Sci.* **664**, 120326 (2024). <https://doi.org/10.1016/j.ins.2024.120326>
19. Yu, X., Xu, Z.: Graph-based multi-agent decision making. *Int. J. Approx. Reason.* **53**, 502–512 (2012). <https://doi.org/10.1016/j.ijar.2011.12.002>
20. Akram, M., Dar, J.M., Naz, S.: Certain graphs under Pythagorean fuzzy environment. *Complex Intell. Syst.* (2019). <https://doi.org/10.1007/s40747-018-0089-5>

21. Naz, S., Akram, M.: Novel decision-making approach based on hesitant fuzzy sets and graph theory. *Comput. Appl. Math.* **38**, 7 (2019). <https://doi.org/10.1007/s40314-019-0773-0>
22. Luqman, A., Akram, M., Alcantud, J.C.R.: Digraph and matrix approach for risk evaluations under Pythagorean fuzzy information. *Expert Syst. Appl.* **170**, 114518 (2021). <https://doi.org/10.1016/j.eswa.2020.114518>
23. Meng, Z., Lin, R., Wu, B.: Multi-criteria group decision making based on graph neural networks in Pythagorean fuzzy environment. *Expert Syst. Appl.* **242**, 122803 (2024). <https://doi.org/10.1016/j.eswa.2023.122803>
24. Zhang, H., Wei, G.: Location selection of electric vehicles charging stations by using the spherical fuzzy CPT-CoCoSo and D-CRITIC method. *Comput. Appl. Math.* (2023). <https://doi.org/10.1007/s40314-022-02183-9>
25. Mishra, A.R., Chen, S.-M., Rani, P.: Multicriteria decision making based on novel score function of Fermatean fuzzy numbers, the CRITIC method, and the GLDS method. *Inf. Sci.* **623**, 915–931 (2023). <https://doi.org/10.1016/j.ins.2022.12.031>
26. Diakoulaki, D., Mavrotas, G., Papayannakis, L.: Determining objective weights in multiple criteria problems: the CRITIC method. *Comput. Oper. Res.* **22**, 763–770 (1995). [https://doi.org/10.1016/0305-0548\(94\)00059-H](https://doi.org/10.1016/0305-0548(94)00059-H)
27. Haktanir, E., Kahraman, C.: A novel picture fuzzy CRITIC & REGIME methodology: wearable health technology application. *Eng. Appl. Artif. Intell.* **113**, 104942 (2022). <https://doi.org/10.1016/j.engappai.2022.104942>
28. Negara, R.M., Syambas, N.R., Mulyana, E.: C3CPS: CRITIC-CoCoSo-based caching placement strategy using multi-criteria decision method for efficient content distribution in named data networking. *J. King Saud Univ. Comput. Inf. Sci.* **35**, 101714 (2023). <https://doi.org/10.1016/j.jksuci.2023.101714>
29. Farid, H.M.A., Dabic-Miletic, S., Riaz, M., Simic, V., Pamucar, D.: Prioritization of sustainable approaches for smart waste management of automotive fuel cells of road freight vehicles using the q-rung orthopair fuzzy CRITIC-EDAS method. *Inf. Sci.* **661**, 120162 (2024). <https://doi.org/10.1016/j.ins.2024.120162>
30. Sun, G., Luo, J., Wang, G.: TOPSIS method based on hesitant factor and priority weighted operator in Pythagorean fuzzy environment. *Int. J. Fuzzy Syst.* **25**, 831–850 (2022). <https://doi.org/10.1007/s40815-022-01406-9>
31. Jin, C., Mi, J., Li, F., Liang, M.: An improved TOPSIS method for multi-criteria decision making based on hesitant fuzzy β neighborhood. *Artif. Intell. Rev.* **56**, 793–831 (2023). <https://doi.org/10.1007/s10462-023-10510-7>
32. Hu, M., Sun, D., Sun, G., Sun, Y., Ouyang, J.: Performance study on anti-weather aging combinations for high-content polymer modified asphalt and comparison by improved multi-scale mathematical TOPSIS method. *Constr. Build. Mater.* **407**, 133357 (2023). <https://doi.org/10.1016/j.conbuildmat.2023.133357>
33. Thilagavathy, A., Mohanaselvi, S.: T-Spherical fuzzy TOPSIS method based on distance measures and Hamacher Heronian mean averaging aggregation operators and its application to waste management. *Appl. Soft Comput.* **162**, 111868 (2024). <https://doi.org/10.1016/j.asoc.2024.111868>
34. Dang, P.-F., Sun, L.-F., Yang, Z.-X., Gong, B., Zhu, J., Yuan, S.: A comprehensive risk assessment method of hydrogenation units integrating DEMATEL-TOPSIS-CM. *Int. J. Hydrogen Energy* **79**, 411–422 (2024). <https://doi.org/10.1016/j.ijhydene.2024.01.094>
35. Xing, Z., Fang, D., Huang, J., Zhao, Y., Wang, L., Wang, J., Zhang, R.: Switching gears: Evaluating Grid-to-Vehicle (G2V) and battery swap (BS) behaviors in new energy vehicles (NEVs) within a flexible institutional landscape. *Travel Behavi. Soc.* **35**, 100739 (2024). <https://doi.org/10.1016/j.tbs.2024.100739>
36. Gurmani, S.H., Zhang, S., Awwad, F.A., Ismail, E.A.A.: Combinative distance-based assessment method using linguistic t-spherical fuzzy aggregation operators and its application to multi-attribute group decision-making. *Eng. Appl. Artif. Intell.* **133**, 108165 (2024). <https://doi.org/10.1016/j.engappai.2024.108165>
37. Zhang, X., Xu, Z.: Extension of TOPSIS to multiple criteria decision making with Pythagorean fuzzy sets. *Int. J. Intell. Syst.* **29**, 1061–1078 (2014). <https://doi.org/10.1002/int.21676>
38. Peng, X., Yang, Y.: Some results for Pythagorean fuzzy sets. *Int. J. Intell. Syst.* **30**, 1133–1160 (2015). <https://doi.org/10.1002/int.21738>
39. Kumar, K., Chen, S.-M.: Group decision making based on entropy measure of Pythagorean fuzzy sets and Pythagorean fuzzy weighted arithmetic mean aggregation operator of Pythagorean fuzzy numbers. *Inf. Sci.* **624**, 361–377 (2023). <https://doi.org/10.1016/j.ins.2022.12.064>
40. Abolghasemi, R., Viedma, E.H., Engelstad, P., Djenouri, Y., Yazidi, A.: A graph neural approach for group recommendation system based on pairwise preferences. *Inf. Fusion* **107**, 102343 (2024). <https://doi.org/10.1016/j.inffus.2024.102343>
41. Ecer, F.: A consolidated MCDM framework for performance assessment of battery electric vehicles based on ranking strategies. *Renew. Sustain. Energy Rev.* **143**, 110916 (2021). <https://doi.org/10.1016/j.rser.2021.110916>
42. Du, J., Meng, X., Li, J., Wu, X., Song, Z., Ouyang, M.: Insights into the characteristics of technologies and industrialization for plug-in electric cars in china. *Energy* **164**, 910–924 (2018). <https://doi.org/10.1016/j.energy.2018.09.060>
43. Zhu, M., Liu, Z., Li, J., Zhu, S.X.: Electric vehicle battery capacity allocation and recycling with downstream competition. *Eur. J. Oper. Res.* **283**, 365–379 (2020). <https://doi.org/10.1016/j.ejor.2019.10.040>
44. Yang, Y., Zhang, Y., Tian, J., Li, T.: Adaptive real-time optimal energy management strategy for extender range electric vehicle. *Energy* **197**, 117237 (2020). <https://doi.org/10.1016/j.energy.2020.117237>
45. Peng, X., Zhang, X., Luo, Z.: Pythagorean fuzzy MCDM method based on CoCoSo and CRITIC with score function for 5G industry evaluation. *Artif. Intell. Rev.* **53**, 3813–3847 (2020). <https://doi.org/10.1007/s10462-019-09780-x>
46. Sarkar, B., Biswas, A.: Pythagorean fuzzy AHP-TOPSIS integrated approach for transportation management through a new distance measure. *Soft. Comput.* **25**, 4073–4089 (2021). <https://doi.org/10.1007/s00500-020-05433-2>
47. Rani, P., Chen, S.-M., Mishra, A.R.: Multi-attribute decision-making based on similarity measure between picture fuzzy sets and the marcos method. *Inf. Sci.* **658**, 119990 (2024). <https://doi.org/10.1016/j.ins.2023.119990>

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.



Jianping Fan received the M.S. and Ph. D. degrees in management from Shanxi University, Taiyuan, China. He is currently a Professor with the School of Economics and Management, Shanxi University. His current research interests include decision forecasting and evaluation.



Yanlong Han received the B.S. degree in Engineering from North University of China and is currently pursuing the M.S. degree with the School of Economics and Management, Shanxi University. His research interests include multi attribute group decision making and machine learning.



Meiqin Wu received the Ph.D. degree in management from Shanxi University, Taiyuan, China. She is currently a Master Tutor in industrial engineering. Her current research interests include decision forecasting and evaluation.